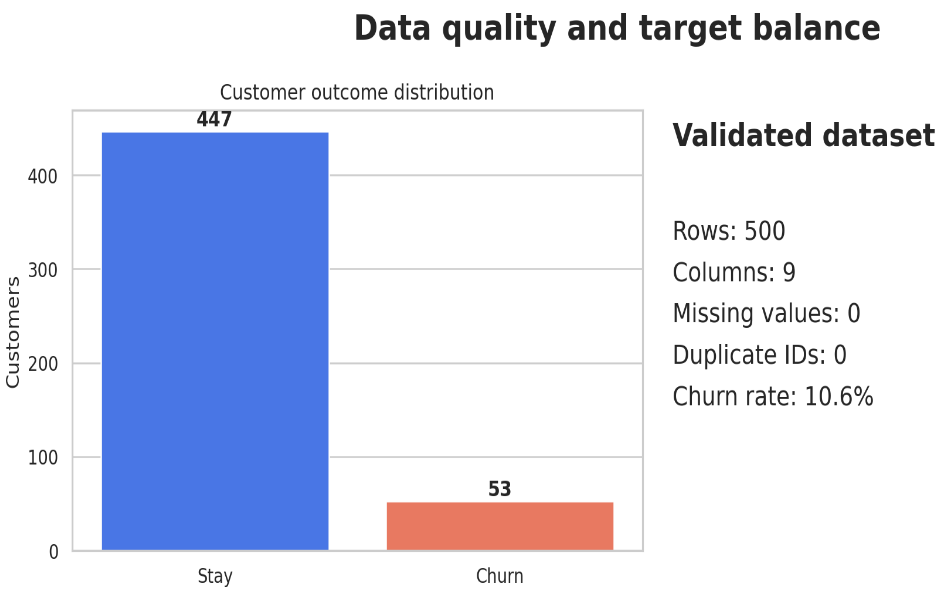
TECHNICAL DOCUMENTATION

Customer Churn Prediction & Retention Strategy

An end-to-end machine learning capstone covering validation, analysis, feature engineering, model tuning, evaluation, interpretation, testing, and deployment.



Prepared for Week 12 Final Capstone & Career Preparation

1. Problem framing and success metrics

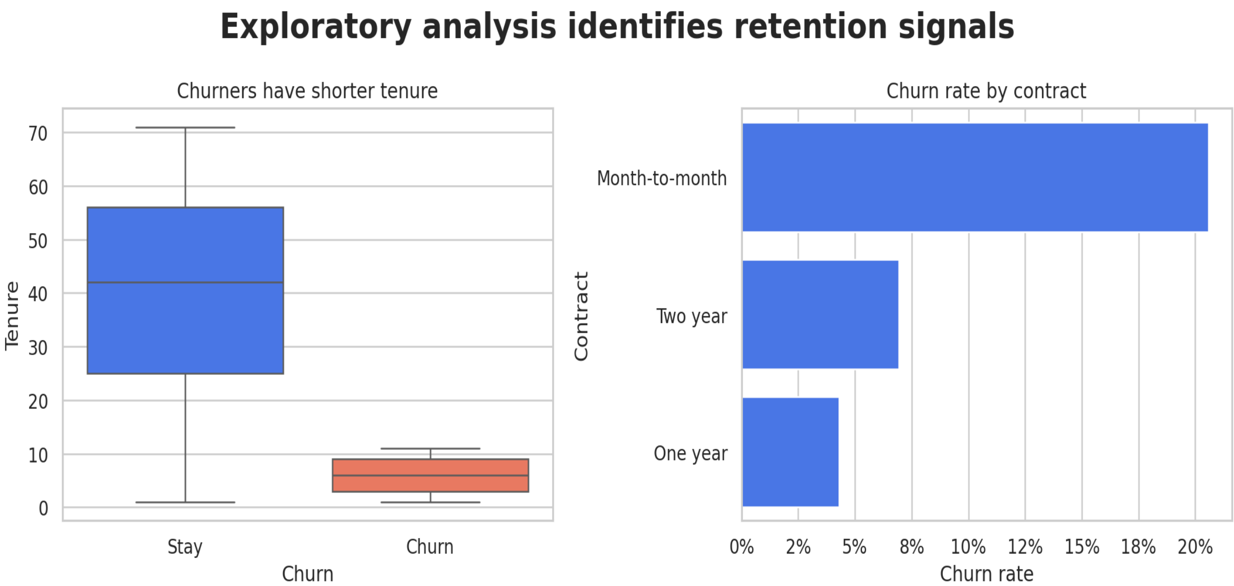
The system estimates churn probability and converts it into an operational risk band. The business goal is to focus limited retention capacity on customers most likely to leave while reducing unnecessary blanket offers.

Declared success thresholds

Metric	Target	Holdout result	Status
Recall	>= 0.80	1.000	Met
ROC-AUC	>= 0.85	1.000	Met
F1	>= 0.65	0.867	Met
Precision	>= 0.50	0.765	Met

Data contract

The validated raw snapshot contains 500 rows, 9 columns, no missing values, unique customer IDs, and 10.6% churn. All observed churn cases have tenure of 12 months or less, an unusually strong pattern that may not generalize. CustomerID is excluded from training. A SHA-256 digest and library versions are recorded with the model.



2. Architecture and preprocessing

Workflow: raw CSV -> validation -> stratified split -> in-pipeline feature engineering -> numeric and categorical preprocessing -> model comparison -> Random Forest GridSearchCV -> training-only threshold selection -> untouched holdout evaluation -> persisted artifact -> API and web app.

Pipeline design

- Schema, range, uniqueness, and missing-value validation occurs before training.
- A 75/25 stratified split preserves the minority class and isolates the final test set.
- Eight business-motivated features are created inside a scikit-learn transformer.
- Numeric variables use median imputation and standardization; nominal categories use most-frequent imputation and one-hot encoding.
- CustomerID and Churn are never input features.

Engineered features

Feature	Purpose
AverageMonthlySpend	Normalizes cumulative charges by tenure
ChargesGap	Compares current charge with historical average
EstimatedAnnualCharges	Annualized current spend
TenureSquared	Captures nonlinear tenure effect
TenureMonthlyInteraction	Combines relationship length and current spend
IsMonthToMonth	Flags flexible contract exposure
IsAutoPay	Flags automated payment channels
IsNewCustomer	Marks first-year customers

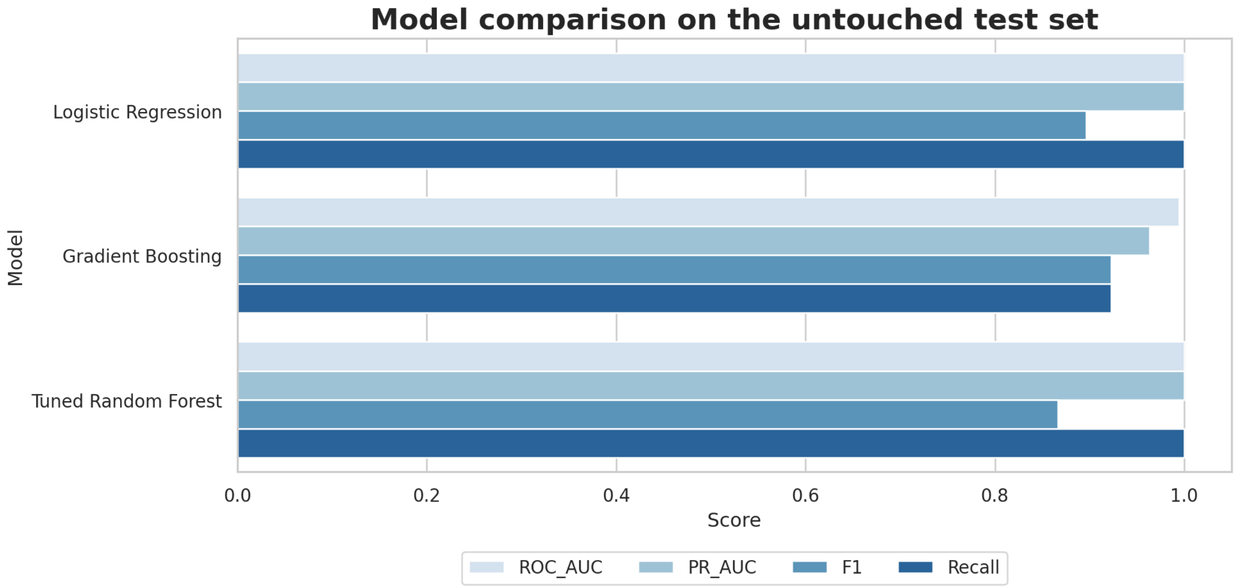
3. Model development and tuning

Logistic Regression and Gradient Boosting establish baseline performance. Random Forest is tuned with five-fold stratified cross-validation across estimator count, maximum depth, minimum leaf size, and maximum features. ROC-AUC is the refit score.

Best cross-validation ROC-AUC: 0.9937
Best parameters: {"model__max_depth": 6, "model__max_features": "sqrt", "model__min_samples_leaf": 2, "model__n_estimators": 300}

Decision threshold

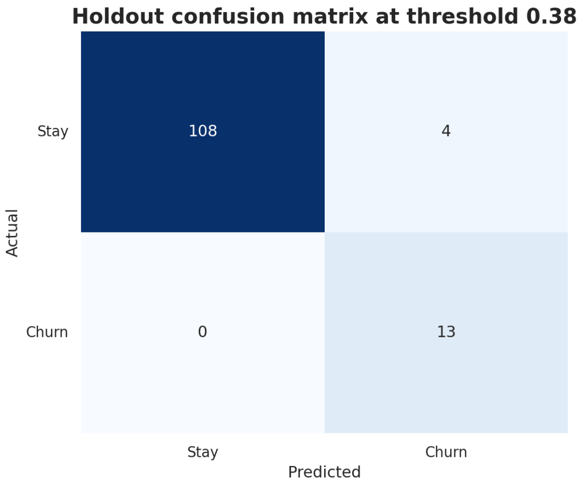
The operating threshold is 0.38. It is selected only from out-of-fold probabilities on the training set using a recall/F1-weighted business score. The test set is not used to select the threshold.

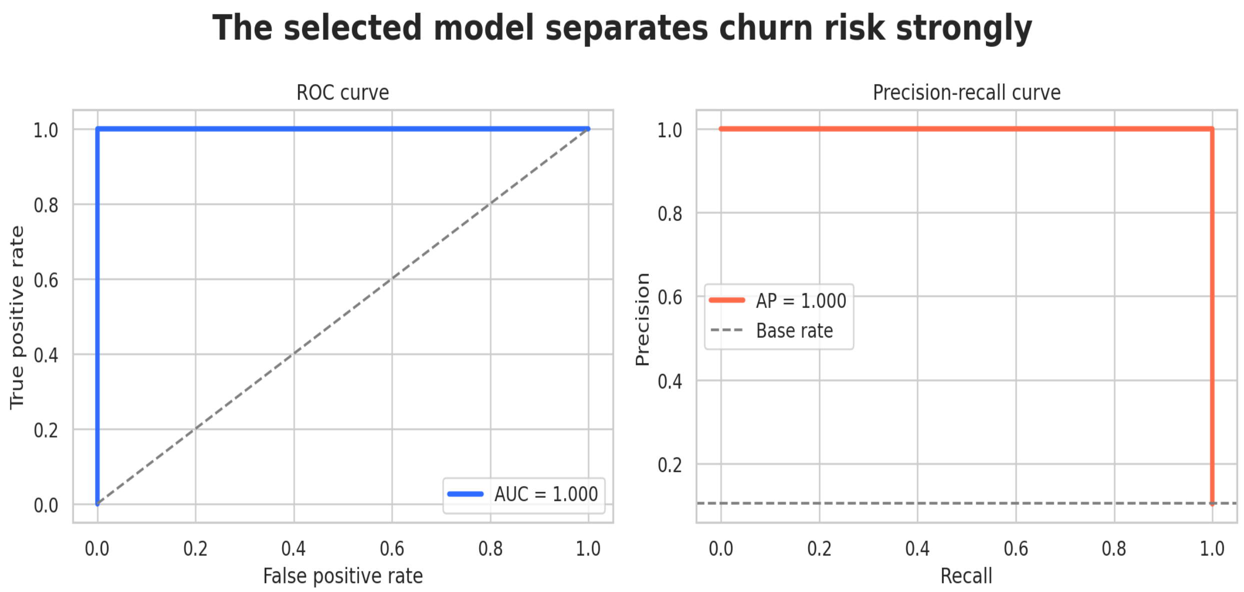


4. Holdout evaluation

Accuracy	Balanced Acc.	Precision	Recall	F1	ROC-AUC	PR-AUC
0.968	0.982	0.765	1.000	0.867	1.000	1.000

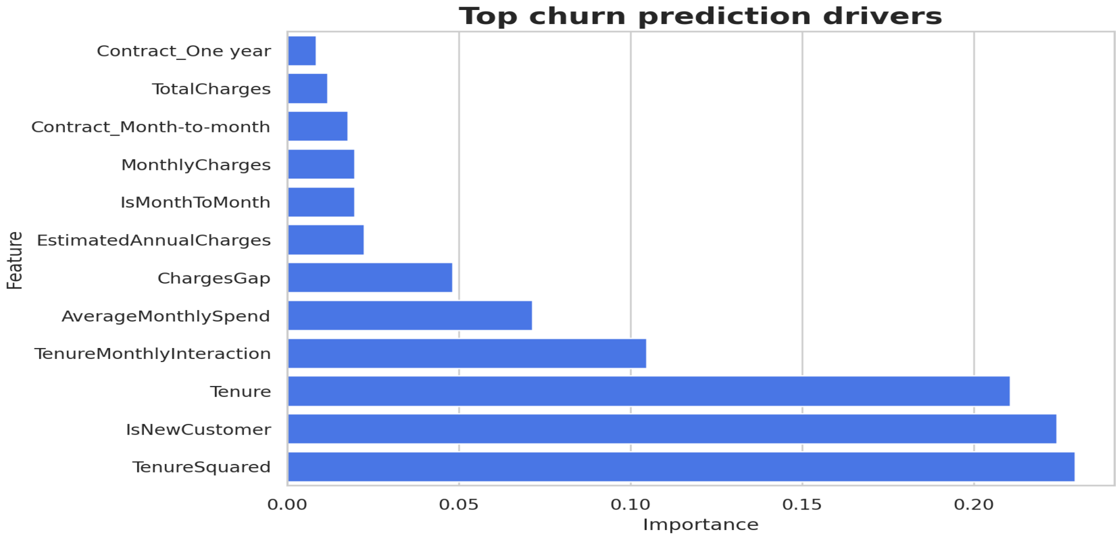
Confusion matrix counts: TN 108, FP 4, FN 0, TP 13. The untouched holdout set contains 125 records.





5. Interpretation and business translation

Global feature importance indicates which transformed inputs the Random Forest used most often. It does not identify causal drivers. Retention offers require controlled testing.



Operational policy

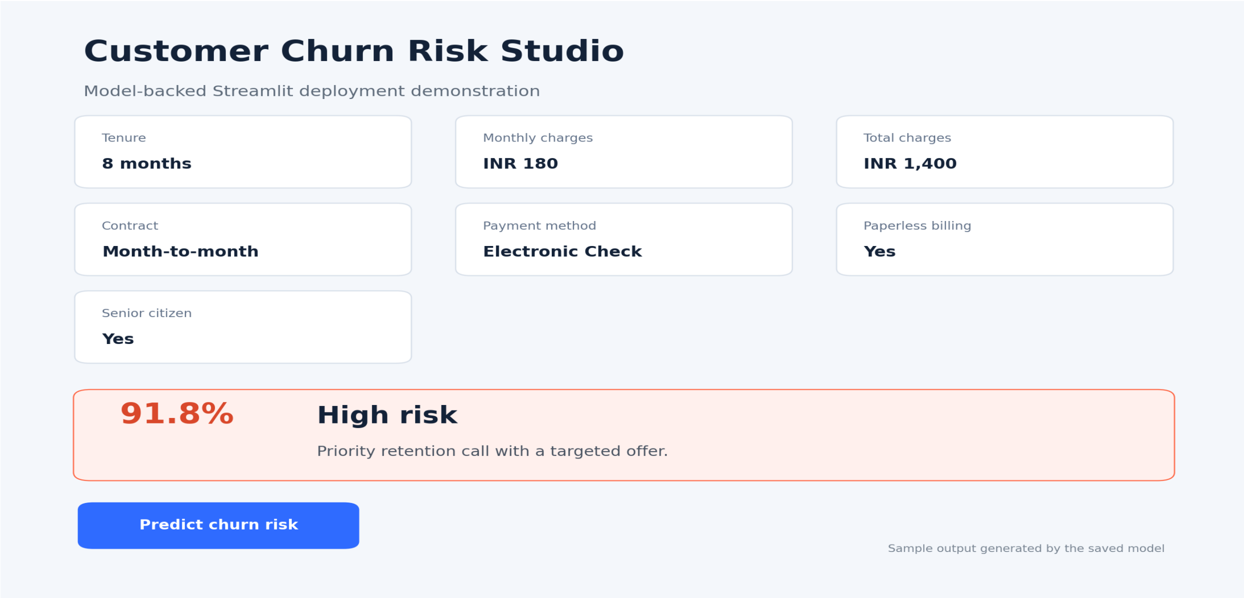
• High risk: priority human review and targeted retention offer.

• Medium risk: proactive support and contract-upgrade message.

• Low risk: regular engagement without unnecessary discount.

6. Deployment architecture

The persisted joblib package includes the complete pipeline, selected threshold, raw feature contract, and risk boundaries. The same prediction function is shared by the CLI, FastAPI service, and Streamlit interface.



Run locally

Streamlit: **streamlit run deployment/streamlit_app.py**
FastAPI: **uvicorn deployment.api:app --reload --port 8000**

Security and reliability

• Load only trusted model artifacts and reproduce the training environment.

• Add authentication, HTTPS, rate limits, structured logs, and secrets management.

• Validate all inputs and provide human review for interventions.

7. Testing, monitoring, and limitations

Automated evidence covers data integrity, split isolation, target leakage prevention, artifact loading, declared performance, repeatable prediction, API validation, executed notebook state, PDF/deck validity, screenshots, and impact assumptions.

Monitoring plan

• Input drift: feature distributions, invalid/missing rates, new category frequency.

• Output drift: risk-band share and score distribution.

• Performance: delayed-label recall, precision, F1, ROC-AUC, PR-AUC, and calibration.

• Business: contact rate, save rate, offer cost, complaints, and incremental lift.

• Fairness: subgroup error rates and intervention burden.

Limitations

• Small educational dataset with no production sampling guarantee; every observed churner has tenure of 12 months or less.

• Random split does not measure future temporal drift.

• Feature importance is not causal and retention uplift is untested.

• Impact values are illustrative assumptions, not realized revenue.

References

Scikit-learn model persistence: https://scikit-learn.org/stable/model_persistence.html
Streamlit deployment: https://docs.streamlit.io/deploy/streamlit-community-cloud/deploy-your-app